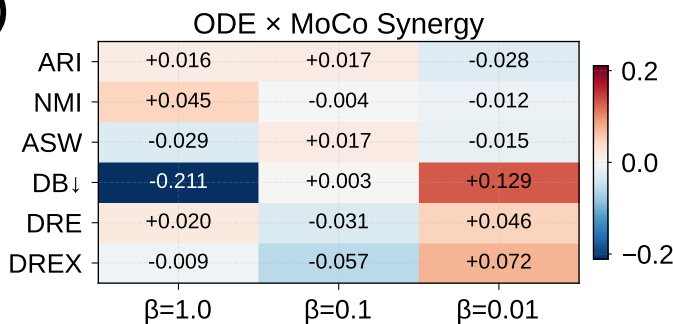


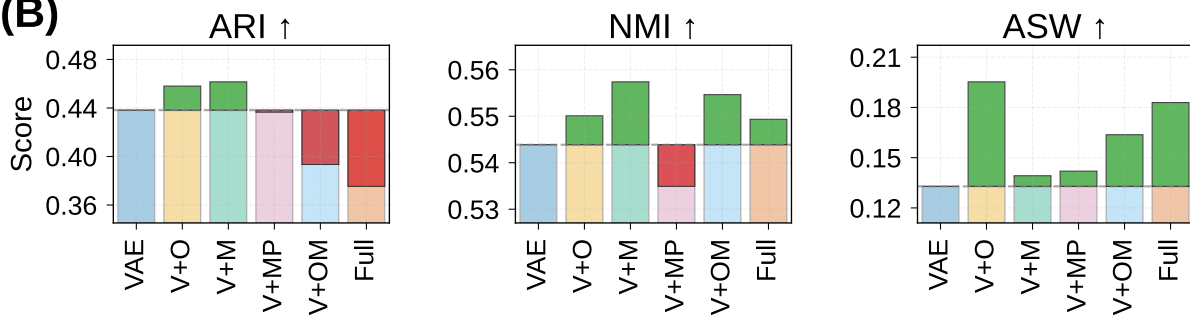
(A)



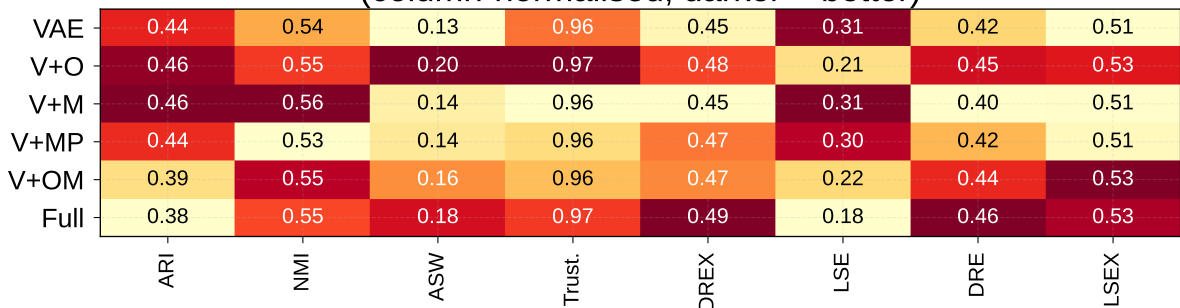
Summary Table

Config	ARI↑	NMI↑	ASW↑	Time(s)	Mem(GB)
VAE	0.438	0.544	0.133	14.800	0.050
V+O	0.458	0.550	0.195	216.700	0.060
V+M	0.462	0.557	0.139	28.900	0.060
V+MP	0.437	0.535	0.142	33.100	0.060
V+OM	0.393	0.555	0.164	283.800	0.080
Full	0.375	0.549	0.183	338.300	0.080

(B)

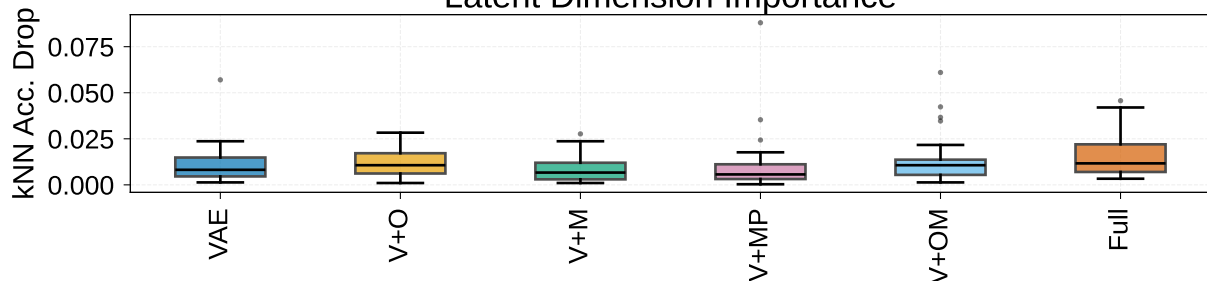


(C)

Performance Heatmap
(column-normalised; darker = better)

(D)

Latent Dimension Importance



ARI = Adj. Rand Index ↑; NMI = Norm. Mutual Info. ↑; ASW = Avg. Silhouette Width ↑; DAV = Davies-Bouldin ↓
 DRE = Dim. Red. Eval. ↑; DREX = Ext. Dim. Red. Eval. ↑; LSE = Latent Space Eval. ↑; LSEX = Ext. Latent Space Eval. ↑
 Config key: V+O = VAE+ODE | V+M = VAE+MoCo | V+MP = VAE+MoCo+Proto | V+OM = VAE+ODE+MoCo